

Experiment No. – 1							
Date of Performance:	28/07/26						
Date of Submission:	29/07/26						
Conceptual Understanding (02)	Execution (1.5)	Problem Solving & Debugging(02)	Professional Conduct & Lab Ethics(02)	Output/ Result Accuracy (1.5)	Documentation & Result Analysis (01)	Experiment Total (10)	Sign with Date

Experiment No. 1 Study control structure statements

1.1 Aim: Study of control structure statement : Write a java program to calculate gross salary & net salary taking the following data. Input: empno, empname, basic salary, DA=70% of basic salary HRA=30% of basic salary CCA=Rs240/- PF=10% of basic salary PT= Rs100/-.

1.2 Course Outcome: Apply fundamental Java programming concepts, including data types, operators, control structures, and functions, to develop basic programs.

1.3 Learning Objectives: Create a simple class based java program to perform specific tasks by building the required logic.

1.4 Requirement: Java development kit (JDK as per OS)

1.5 Related Theory:

The Java Statements which control the flow of execution of the program are called Control Structure or Control Flow Statements. There are three types of control flow statements:

Decision Making:

- If Statement
- If Else Statement
- Switch Statement

Loop Statement:

- For Loop
- While Loop
- Do while Loop

Jump Statement:

- Continue
- Break

Decision Making Statements:

If Statements: Execute the block of code if the condition is true.

```
if (condition)
{
    statement 1;
}
```

If else Statements: Execute the block of code if condition is true else execute another block of code.

```
if (condition)
{
    statement 1;
}
else
{
    statement 2;
}
```

Nested If else statement: contain a if or if-else statement inside another if or else-if statement.

```
if(condition 1)
{
    statement 1;
    if(condition 2)
    {
        statement 2;
    }
    else
    {
        statement 2;
    }
}
```

If else Ladder: if-statement followed by multiple else-if statements

```

if(condition 1)
{
    statement 1;
}
else if(condition 2)
{
    statement 2;
}
else
{
    statement 2;
}

```

Switch Statement: Contains multiple blocks of code also called as cases. Any one block of code will get executed depending on the variable which is being switched. After every case the Break statement must be added to ensure the execution of a particular case only. Default cases can also be added, default cases executed when none of the cases satisfy the condition.

```

switch (expression)
{
    case value1: statement1;
        break;
    case value2: statement2;
        break;

        .
        .
        .
    case valuex: statementx;
        break;

    default: default statement;
}

```

Loop Statements:

For Loop: Block of code will be executed repeatedly depending upon condition.

Syntax 1: -

```

for(initialization ; condition ; increment/decrement)
{
    statement1;
    statement2;
    statement3;
}

```

Syntax 2: -

```

initialization
for( ;condition ; increment/decrement)
{
    statement1;
}

```

```
statement2;
statement3;
}
```

Syntax 3: -

```
initialization;
for(; condition ;)
```

```
{
statement1;
statement2;
statement3;
```

```
increment/decrement;
}
```

Example:

Class ABC

```
{
public static void main(String[] args)
{
int add = 0;
for(int j = 1; j<=10; j++)
{
    add = add+ j;
}
System.out.println("The sum of first 10 numbers is " + add);
}
}
```

While Loop: Block of code will be executed repeatedly till condition remains true. In case of While loop we don't know the number of iterations in advance.

Syntax:

```
while(condition)
```

```
{
statement1;
statement1;
```

```
.
.
.
}
```

Example:

public class Calculation

```
{
public static void main(String[] args)
{
    int i = 0;
```

```

        System.out.println("Printing the list of first 10 even numbers \n");
while(i<=10)
{
    System.out.println(i);
    i = i + 2;
}
}
}
}

```

Do while Loop: Block of code will be executed repeatedly till condition remains true. In the case of the Do While loop also we don't know the number of iterations in advance. In case of Do while loop, irrespective of condition the block of code will be executed at least once as Do while loop exit-controlled loop.

Syntax:

```

do
{
statement1;
statement2;
statement3;
statement4;
}while (condition);

```

Example:

```

public class JPL
{
public static void main(String[] args)
{
    int i = 0;
    System.out.println("Printing the list of first 10 even numbers \n");
    do
    {
        System.out.println(i);
        i = i + 2;
    }while(i<=10);
}
}

```

Jump Statements: used to transfer the control of the program to the specific part of the program.

Break Statements: used to break the flow of the program and transfer the control to the next statement outside a loop or switch statement. In case of nested loops, the Break statement will break the flow of only the inner loop.

Syntax:

```

Class ABC
{
int a;
switch(a){

```

Case 1:

```

        Statement1;
        Break;

    Case 2:
        Statement1;
        Break;
        .
        .
        .
    Default:
        Statement1;
    }
}

```

Example:

```

public class ABC
{
    public static void main(String[] args)
    {
        for(int i = 0; i<= 10; i++)
        {
            System.out.println(i);
            if(i==6) {
                break;
            }
        }
    }
}

```

Continue Statement: Continue statement skips the part of loop after continue statements and jumps to the next iteration of the same loop immediately.

Example:

```

public class ABCcontinue
{
    public static void main(String[] args)
    {
        for(int i = 0; i<= 2; i++)
        {
            for (int j = i; j<=5; j++)
            {
                if(j == 4)
                {
                    continue;
                }
                System.out.println(j);
            }
        }
    }
}

```

1.6 Procedure:

Create a java class with main method, declare variables for data, and apply logic for finding gross salary and net salary where basic salary, DA, HRA, PF and CCA components are provided.

1.7 Program and Output:

Code:

```
import java.util.Scanner;

class EmployeeSalary {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int empNo = sc.nextInt();
        sc.nextLine();
        String empName = sc.nextLine();
        double basic = sc.nextDouble();

        double da = basic * 0.70;
        double hra = basic * 0.30;
        double cca = 240;
        double pf = basic * 0.10;
        double pt = 100;

        double gross = basic + da + hra + cca;
        double net = gross - pf - pt;

        System.out.println("\n----- Employee Salary Details -----");
        System.out.println("Employee Number : " + empNo);
        System.out.println("Employee Name : " + empName);
        System.out.println("Basic Salary : " + basic);
        System.out.println("DA : " + da);
        System.out.println("HRA : " + hra);
        System.out.println("CCA : " + cca);
        System.out.println("PF : " + pf);
        System.out.println("PT : " + pt);
        System.out.println("Gross Salary : " + gross);
        System.out.println("Net Salary : " + net);

        sc.close();
    }
}
```

}

Parikshak Online Program Grading Tool - Personal - Microsoft Edge
https://parikshak.in/onlineTestForm.html?testId=3644288

Welcome sakec_365

सि डैक CDAC प्रगत संगणन विकास केंद्र CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING

Time Left: 19:59:48

Java Experiment 1 | Last attempt code | Recover Code

SYIT1-Java-Experiment1

Problem Statement | Input Specification | Output Specification | Sample I/O | Image

java (1.8) | Filename: EmployeeSalary.java

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```
1 import java.util.Scanner;
2
3 class EmployeeSalary{
4     public static void main(String[] args){
5         Scanner sc = new Scanner(System.in);
6
7         int empNo = sc.nextInt();
8         sc.nextLine();
9         String empName = sc.nextLine();
10        double basic = sc.nextDouble();
11
12        double da = basic * 0.70;
13        double hra = basic * 0.30;
14        double cca = 740;
15        double pf = basic * 0.10;
16        double pt = 100;
17
18        double gross = basic + da + hra + cca;
19        double net = gross - pf - pt;
20
21        System.out.println("Employee Salary Details :-");
22        System.out.println("Employee Number : " + empNo);
23        System.out.println("Employee Name : " + empName);
24        System.out.println("Basic Salary : " + basic);
25        System.out.println("DA : " + da);
26        System.out.println("HRA : " + hra);
27        System.out.println("CCA : " + cca);
28        System.out.println("PF : " + pf);
29        System.out.println("PT : " + pt);
30        System.out.println("Gross Salary : " + gross);
31        System.out.println("Net Salary : " + net);
32
33        sc.close();
34    }
35 }
```

Provide your own test cases:

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Input

```
1 5
2 Janvi
3 50000
```

Ln: 2, Ch: 5, Total Ln: 3

☒ Toggle editor

Compile | Self Assessment | Submit to Grader

Output:

```
Execution Time : 0.05 Second
Execution Space : 4 Kbyte
-----
Y.
-- Accepted --
```

Close Test

1.8 Conclusion: In this experiment, we learned the basics of Java programming, including how to take input from the user using the Scanner class and how to control the flow of a program using control structures and looping statements. We studied the working and syntax of for, while, and do-while loops and understood their appropriate use cases. This experiment helped in developing a

clear understanding of program execution, repetition, and user interaction in Java.

1.9 Questions:

1. What is a control structure statement?

Ans- A control structure statement is a statement that controls the flow of execution of a program. It determines the order in which statements are executed based on conditions or repetition.

Types of control structures:

- Sequential: Executes statements one after another.
- Selection (Decision): if, if-else, switch
- Iteration (Looping): for, while, do-while
- Jump: break, continue, return

2. What are various loops in Java and its syntax?

Ans- There are three types of loops in Java.

a) for loop

Syntax:

```
for(initialization; condition; increment/decrement)
{
    // statements
}
```

b) while loop

Syntax:

```
while(condition)
{
    // statements
}
```

c) do-while loop

Syntax:

do

```
{  
    // statements  
} while(condition);
```

3. State the working of for loop, while loop and do-while loop.

Ans-

For Loop:

- Used when the number of repetitions is known.
- First, it initializes the variable.
- Then it checks the condition.
- If the condition is true, the statements are executed.
- This process repeats until the condition becomes false.

While Loop:

- Used when the number of repetitions is not known.
- It checks the condition first.
- If the condition is true, the loop executes.
- If the condition is false, the loop stops.

Do-While Loop:

- It executes the statements first.
- Then it checks the condition.
- It always runs at least one time, even if the condition is false.

4. How to take input from user in Java programming.

Ans- In Java, user input is taken using the Scanner class.

```
import java.util.Scanner;
```

```
Scanner sc = new Scanner(System.in);
```

```
int num = sc.nextInt();    // Input integer
double salary = sc.nextDouble(); // Input decimal number
String name = sc.nextLine(); // Input string
```